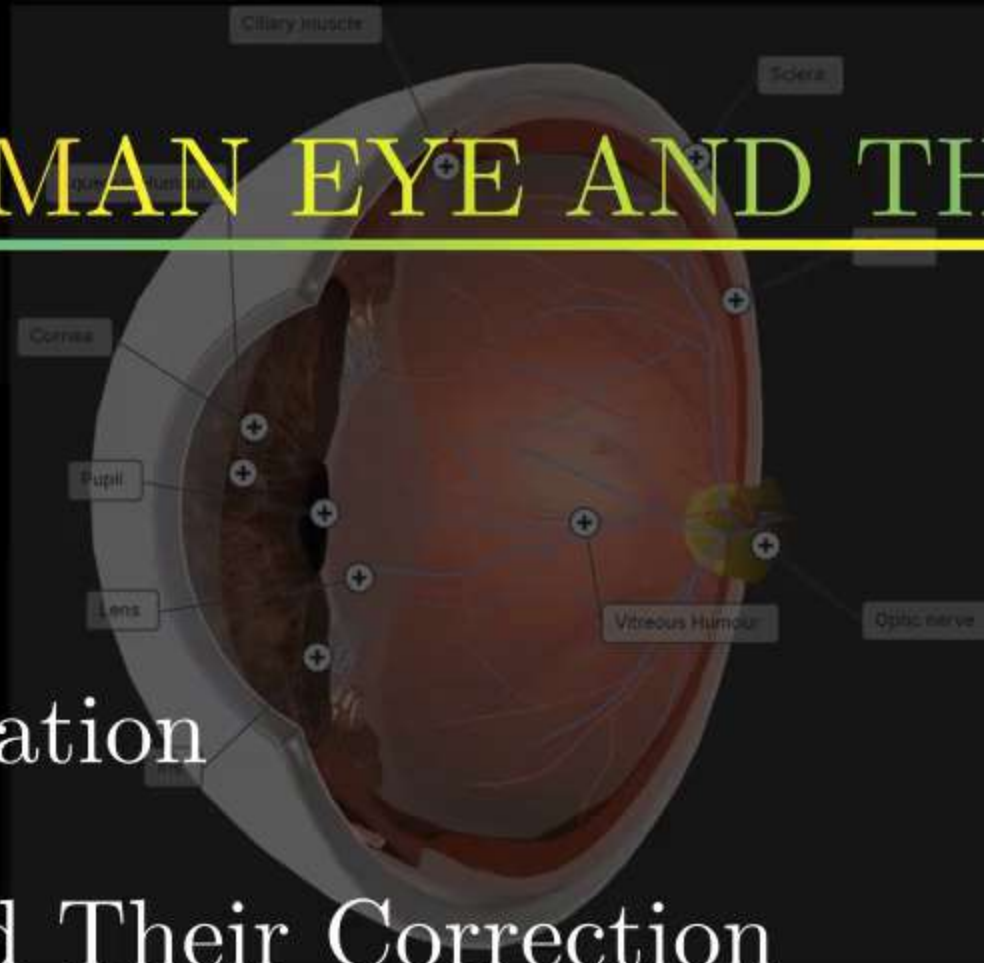


CHAPTER 2 : THE HUMAN EYE AND THE COLOURFUL WORLD

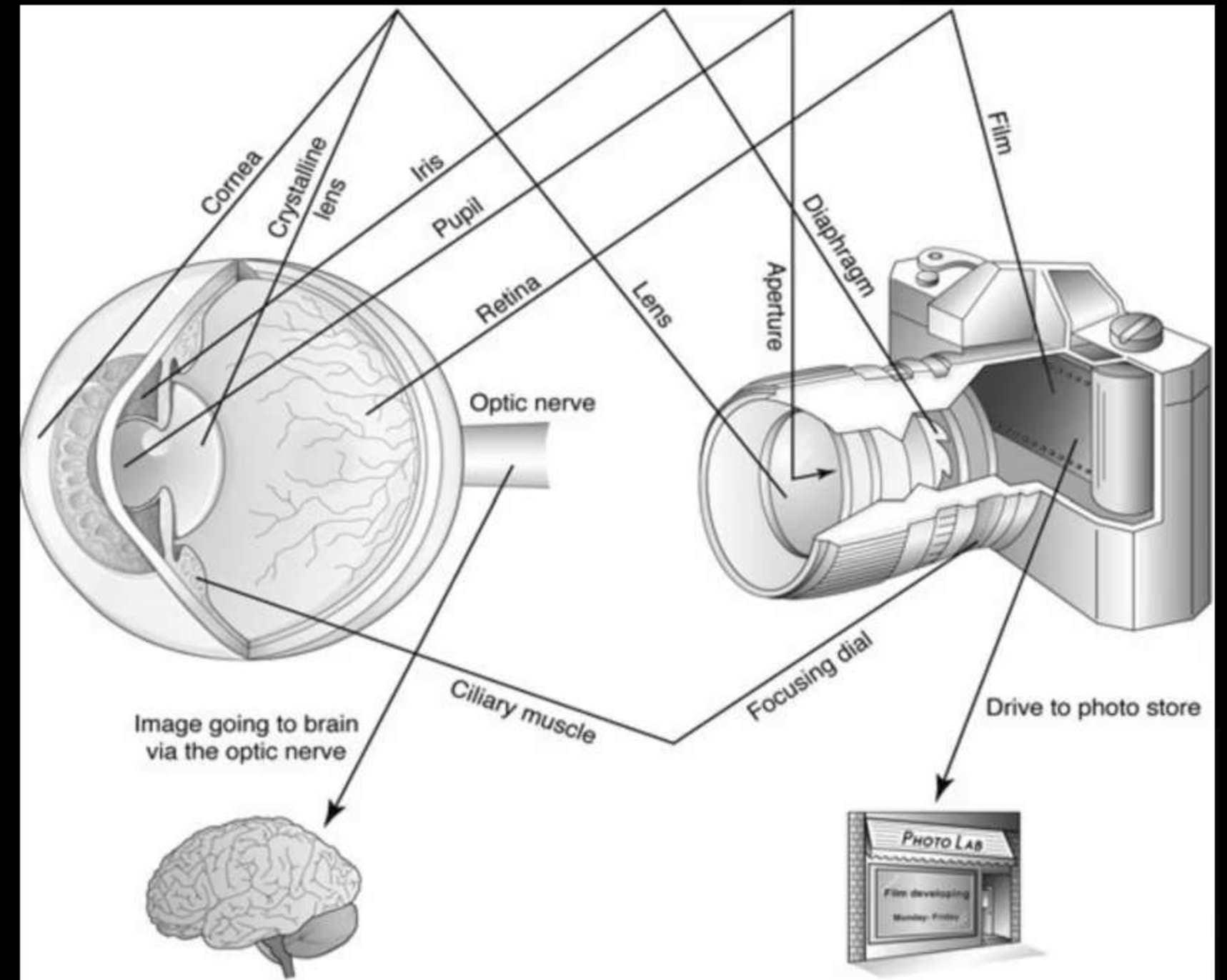
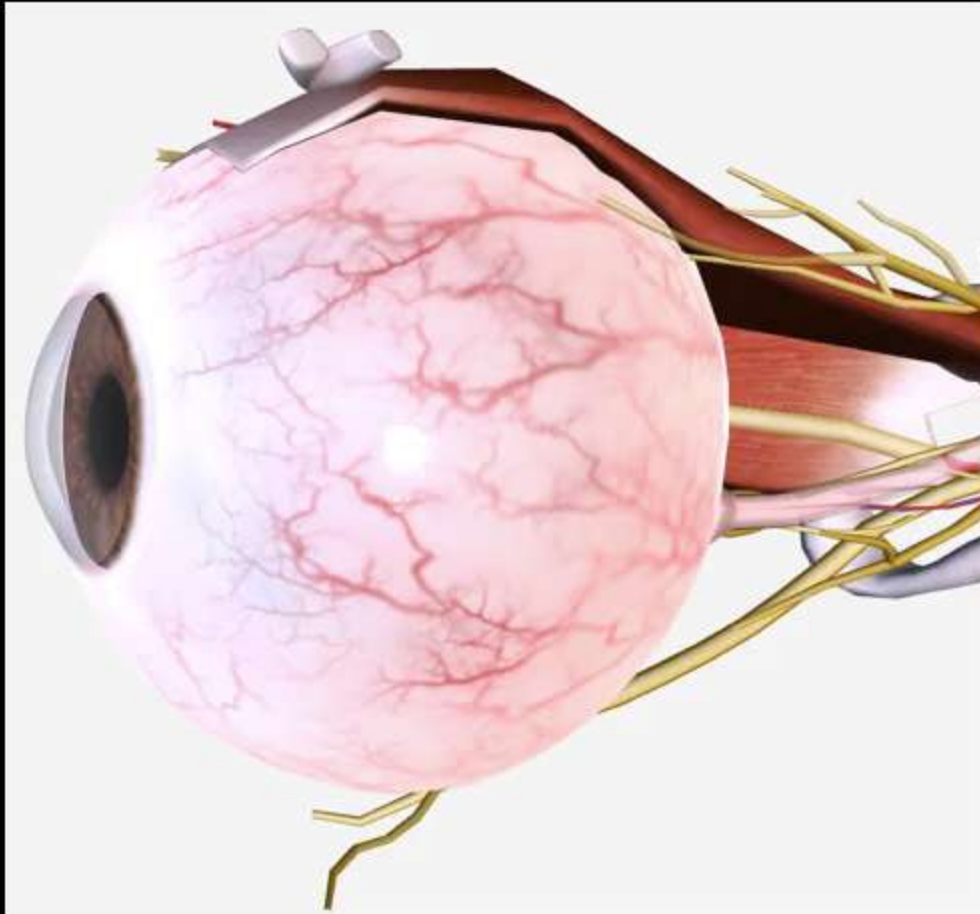
Learning Objectives :

- The Human Eye
- Power of Accommodation
- Defects of Vision and Their Correction
- Refraction of Light Through a Prism
- Dispersion of White Light by a Glass Prism
- Atmospheric Refraction
- Scattering of Light



The Human Eye

- The human **eye** is one of the **most valuable and sensitive sense organs**. It enables us to see the wonderful world and the colours around us.
- The human **eye** is like a **camera**.
- The **eyeball** is **approximately spherical** in shape with a **diameter** of about **2.3 cm**.



Parts and Function of Human Eye

- Cornea :

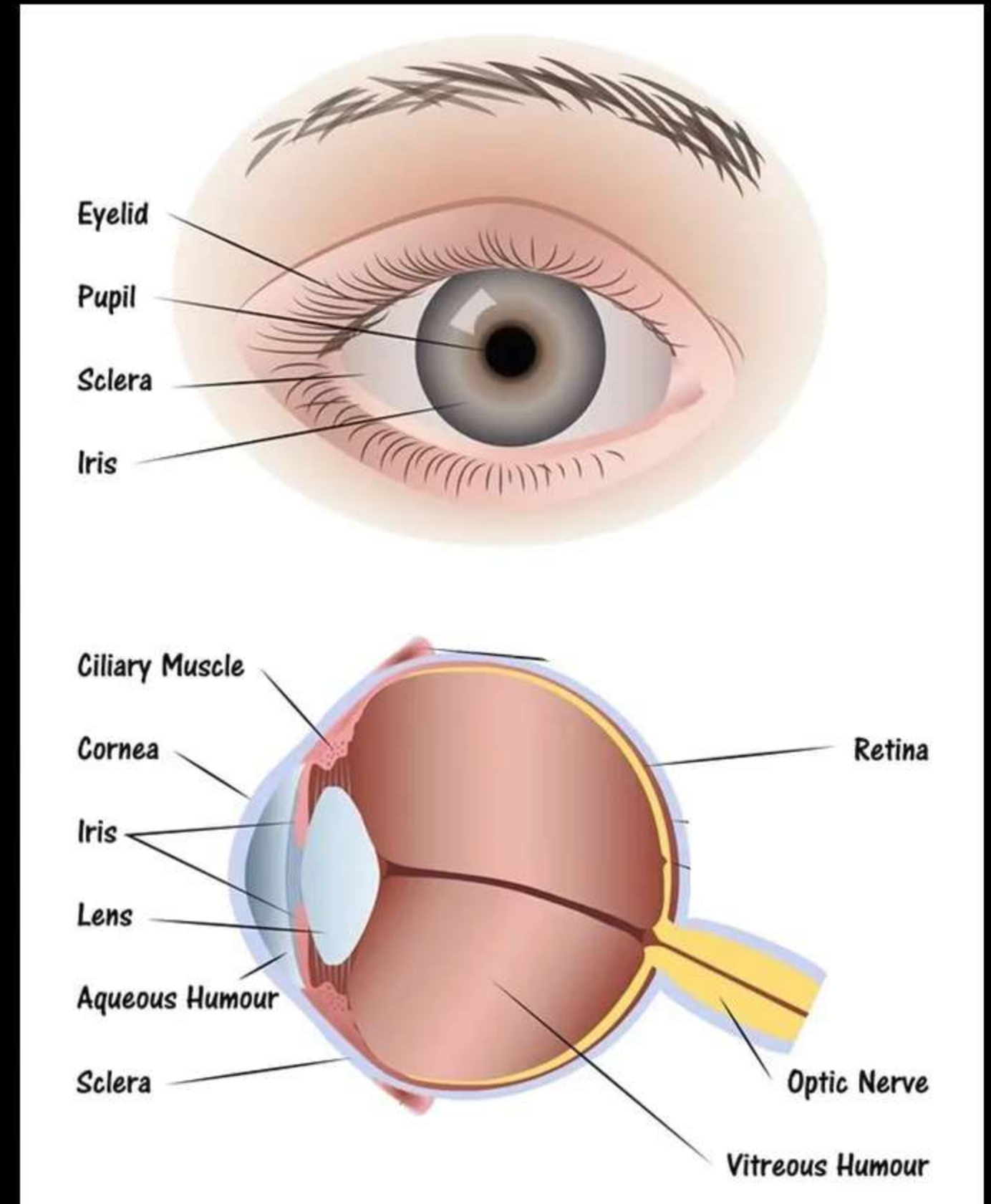
It is a thin transparent bulged membrane on the front of the eyeball. Most of the refraction for the light entering the eye occurs at the outer surface of the cornea.

- Iris :

Iris is a dark muscular diaphragm and coloured part of the eye behind the cornea, that controls the size of the pupil.

- Pupil :

The black, small opening (hole) in the centre of iris. The pupil regulates and controls the amount of light entering the eye.



Parts and Function of Human Eye

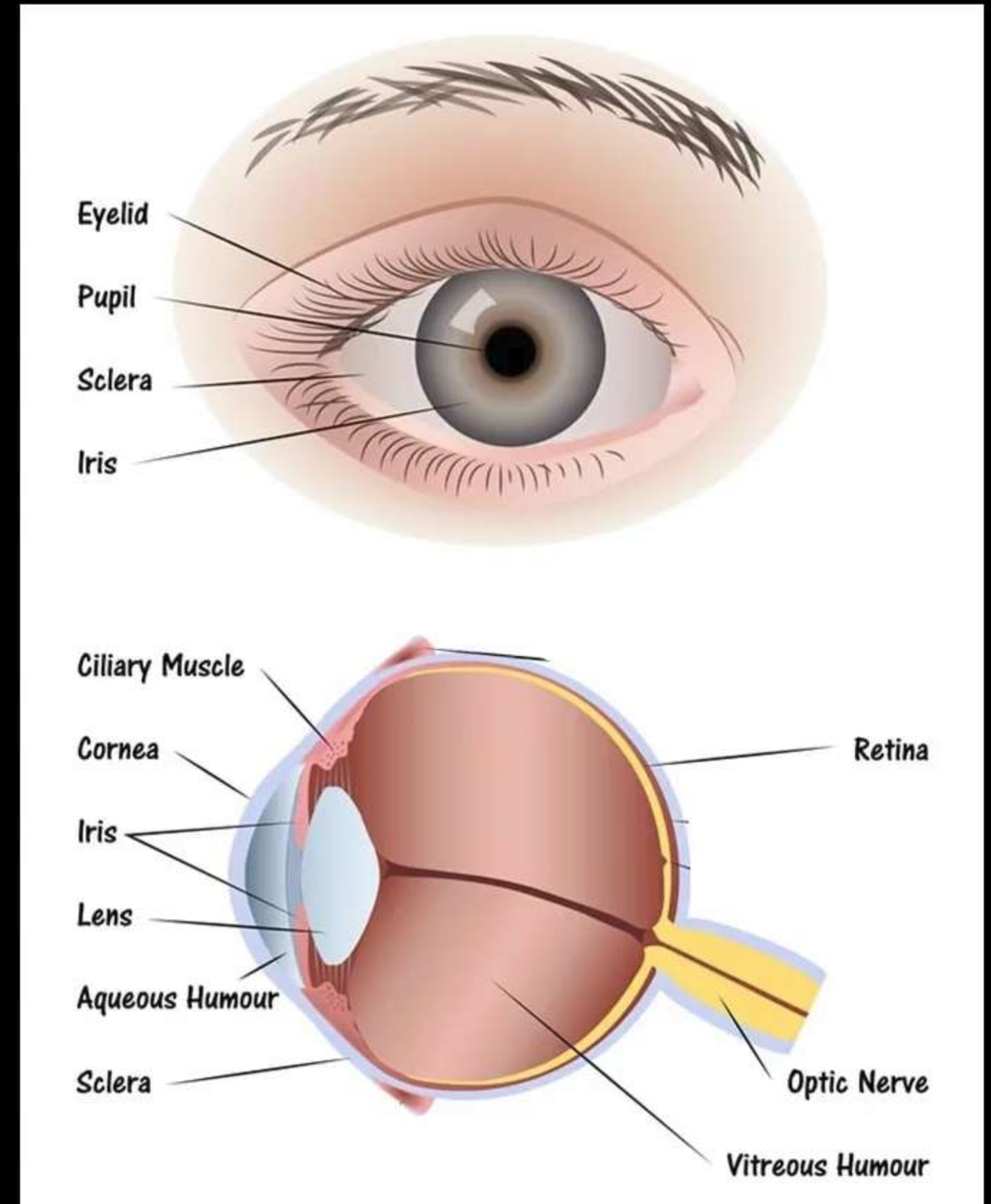
- Crystalline lens :

It is a **biconvex lens**, composed of **fibrous, jelly like material**. It forms an **inverted real image of the object on the retina**.

The **shape/focal length of the lens can be adjusted** with the help of **ciliary muscles** to focus objects at different distances on the retina.

- Retina :

The retina is a **delicate membrane** having enormous number of **light-sensitive cells (Rods and Cones)**. The light-sensitive cells get activated upon illumination and **generate electrical signals**. These signals are sent to the brain via the **optic nerves**



Power of Accommodation

- Accommodation :

The ability of the eye lens to adjust its focal length is called accommodation..

- Role of Ciliary muscles :

The ciliary muscles are attached to the crystalline lens and change its focal length/curvature by contracting and relaxing.

- When the muscles are relaxed, the lens becomes thin. Thus, its focal length increases. This enables us to see distant objects clearly.
- When you are looking at objects closer to the eye, the ciliary muscles contract. This increases the curvature of the eye lens. The eye lens then becomes thicker. Consequently, the focal length of the eye lens decreases
- However, the focal length of the eye lens cannot be decreased below a certain minimum limit.

Power of Accommodation

- Near Point :

The **minimum distance**, at which objects can be seen most **distinctly without strain**, is called the **least distance of distinct vision**. It is also called the **near point of the eye**. For a young adult with normal vision, the near point is about **25 cm**.

- Far Point:

The **farthest point** upto which the eye can **see objects clearly** is called the **far point of the eye**. It is **infinity** for a normal eye.

- Cataract:

Sometimes, the crystalline **lens** of people at **old age** becomes **milky and cloudy**. This condition is called **cataract**. This **causes partial or complete loss of vision**. It is possible to restore vision through a **cataract surgery**.

Myopia (Near-sightedness)

- Myopia :

Myopia is a **defect of vision** in which a person **can see nearby objects clearly** but **cannot see distant objects clearly**. A myopic person has the **far point nearer than infinity**.

- Causes of Myopia :

(i) The eye lens becomes **too curved** (focal length decreases)

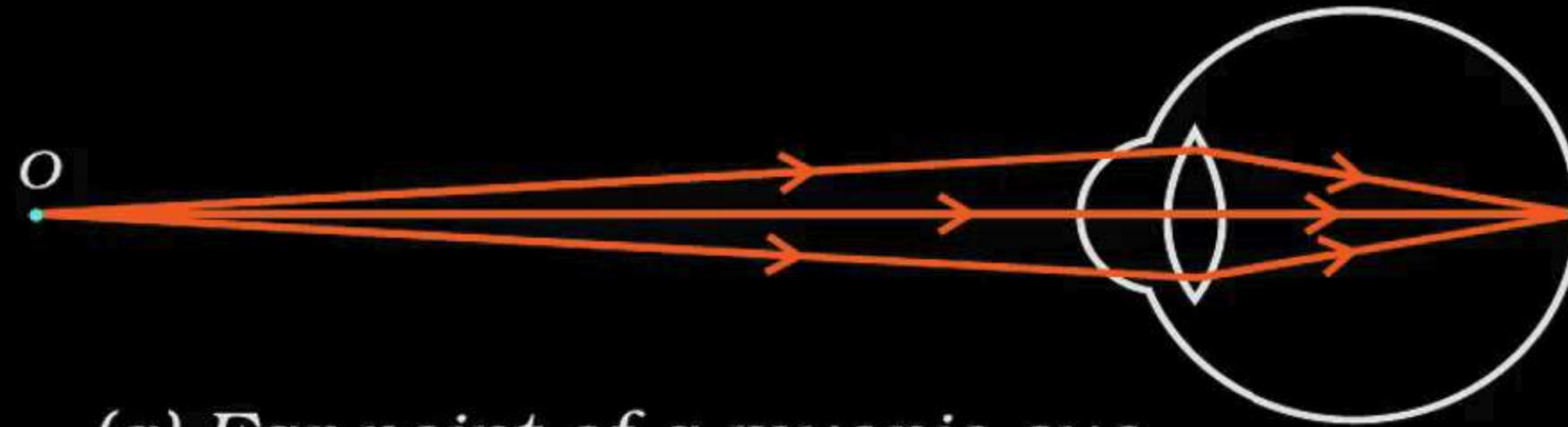
(ii) The eyeball is **elongated (too long)**.

This causes the **image** to be formed in **front of the retina**.

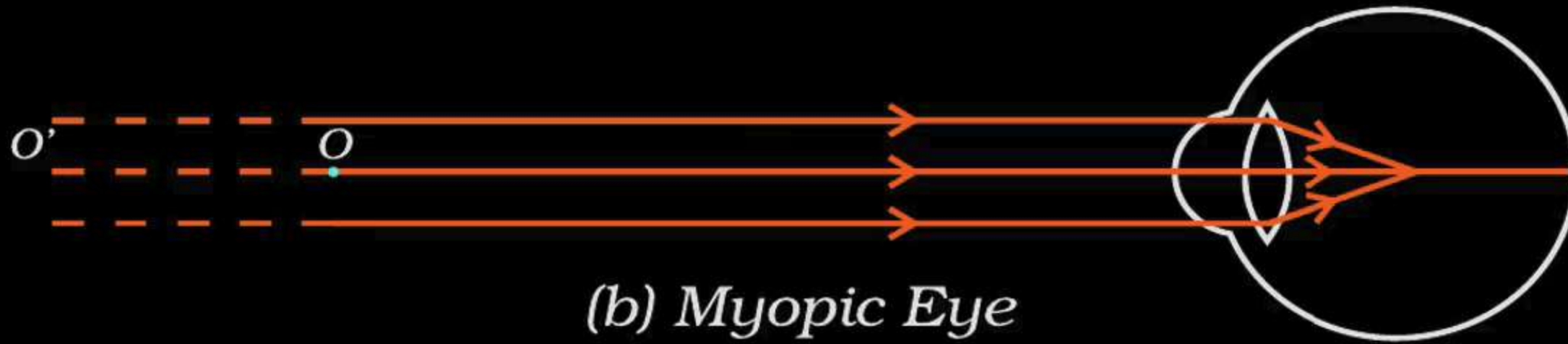
- Correction of Myopia :

Myopia can be corrected by using a **concave lens** of suitable focal length. The concave lens forms an image back on to the retina.

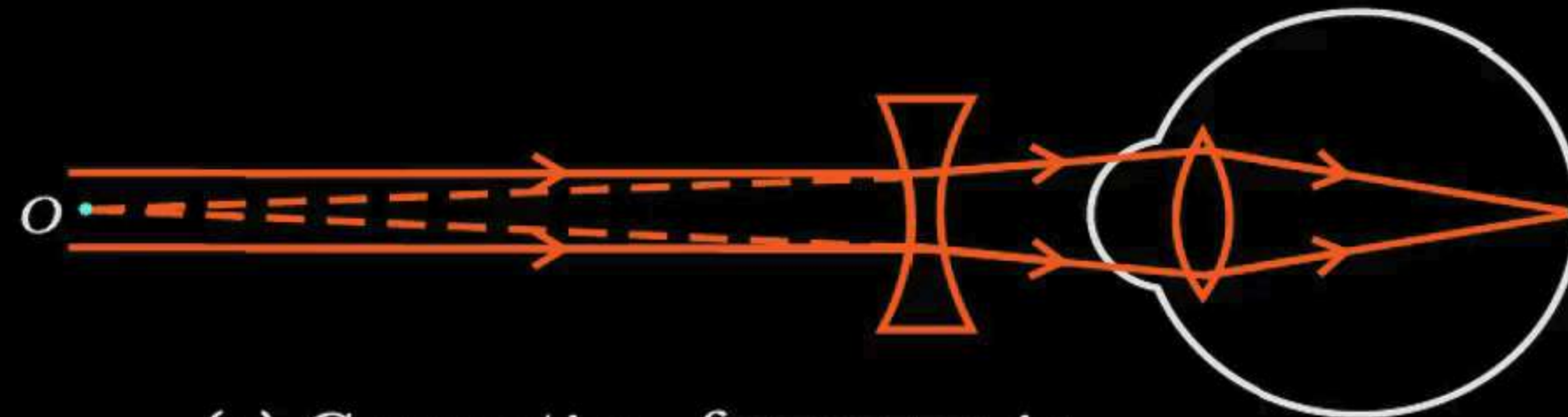
Myopia (Near-sightedness)



(a) Far point of a myopic eye



(b) Myopic Eye



(c) Correction for myopia

Hypermetropia (Far-sightedness)

- Hypermetropia :

Hypermetropia is a defect of vision in which a person can see distant objects clearly but cannot see nearby objects clearly. A hypermetropic person has the near point farther than 25 cm.

- Causes of Hypermetropia :

(i) The eye lens focal length increases)

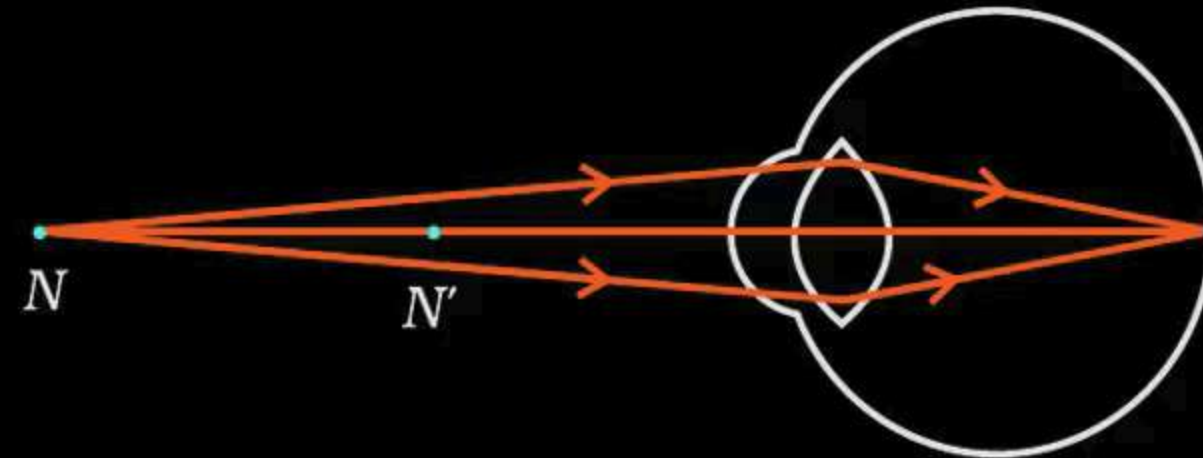
(ii) The eyeball is shortened (too small).

This causes the image to be formed behind the retina.

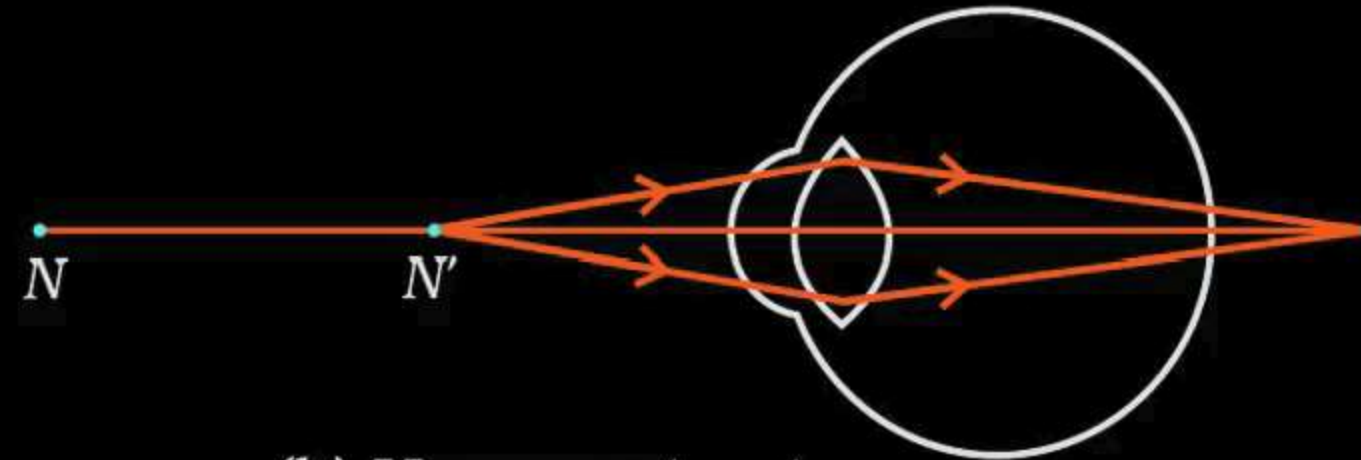
- Correction of Hypermetropia :

Hypermetropia can be corrected by using a **convex lens** of suitable focal length. The convex lens forms an image back on to the retina.

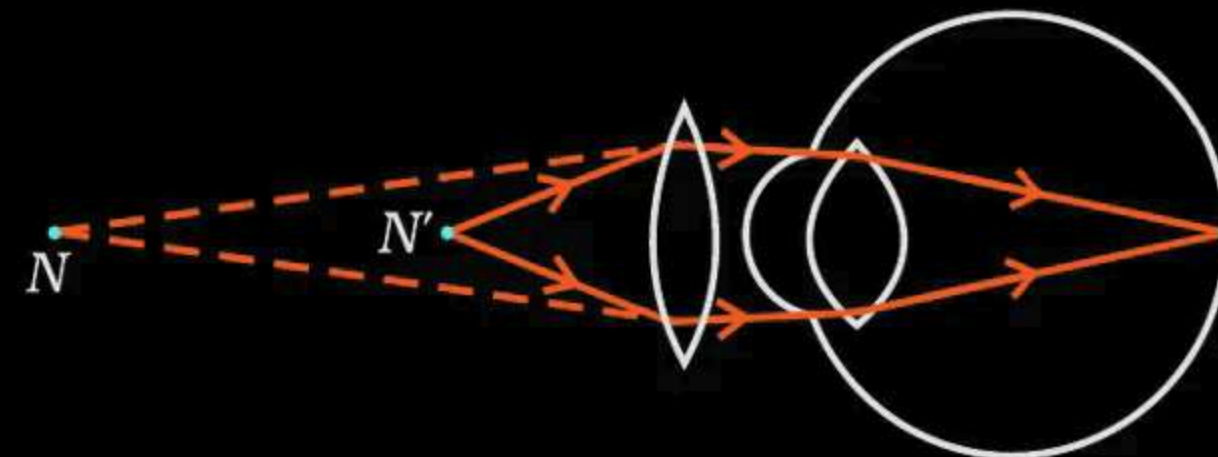
Hypermetropia (Far-sightedness)



(a) Near point of a Hypermetropic eye



(b) Hypermetropic eye



(c) Correction for Hypermetropic eye

Presbyopia

- Presbyopia :

Presbyopia is a defect of vision in which a person cannot see nearby objects clearly and some person can also cannot see distant objects clearly. It is caused due to the gradual weakening of ciliary muscles and diminishing flexibility of the eye lens with age.

- Correction of Presbyopia :

Presbyopia can be corrected by using a bifocal lens consists of both concave and convex lenses. The upper portion consists of a concave lens. It facilitates distant vision. The lower part is a convex lens. It facilitates near vision.

Presbyopia

Bifocal Lens

Concave lens
for Distant vision



Convex lens
for Near vision

Refraction of Light Through a Prism

- Triangular Glass Prism :

It has two triangular bases and three rectangular lateral surfaces. These surfaces are inclined to each other.

- Angle of the prism :

The angle between its two lateral faces is called the angle of the prism.

- Refraction of light by prism :

A ray of light is entering from air to glass at the first surface **AB**. The light ray on refraction has bent towards the normal. At the second surface **AC**, the light ray has entered from glass to air. Hence it has bent away from normal.

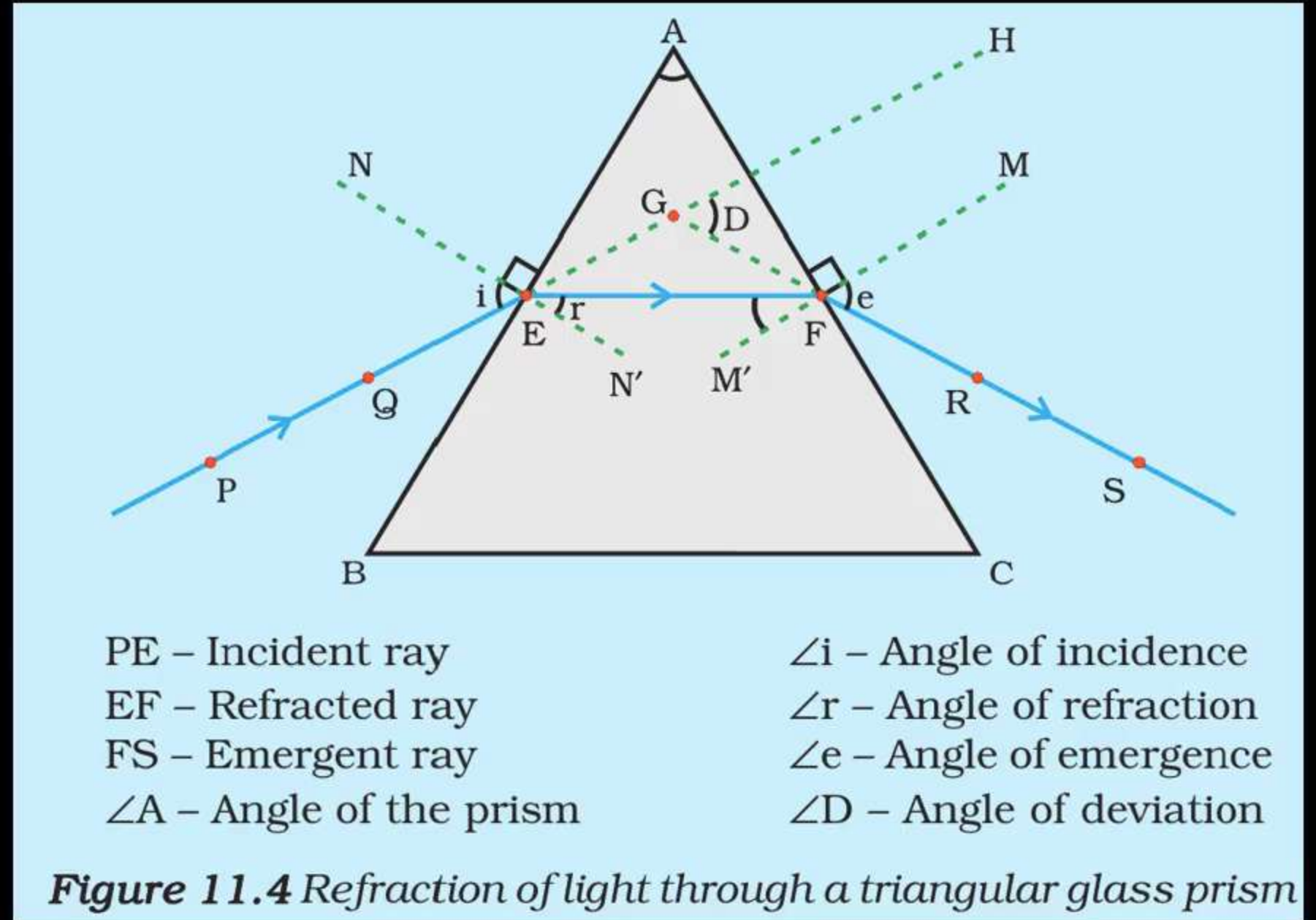
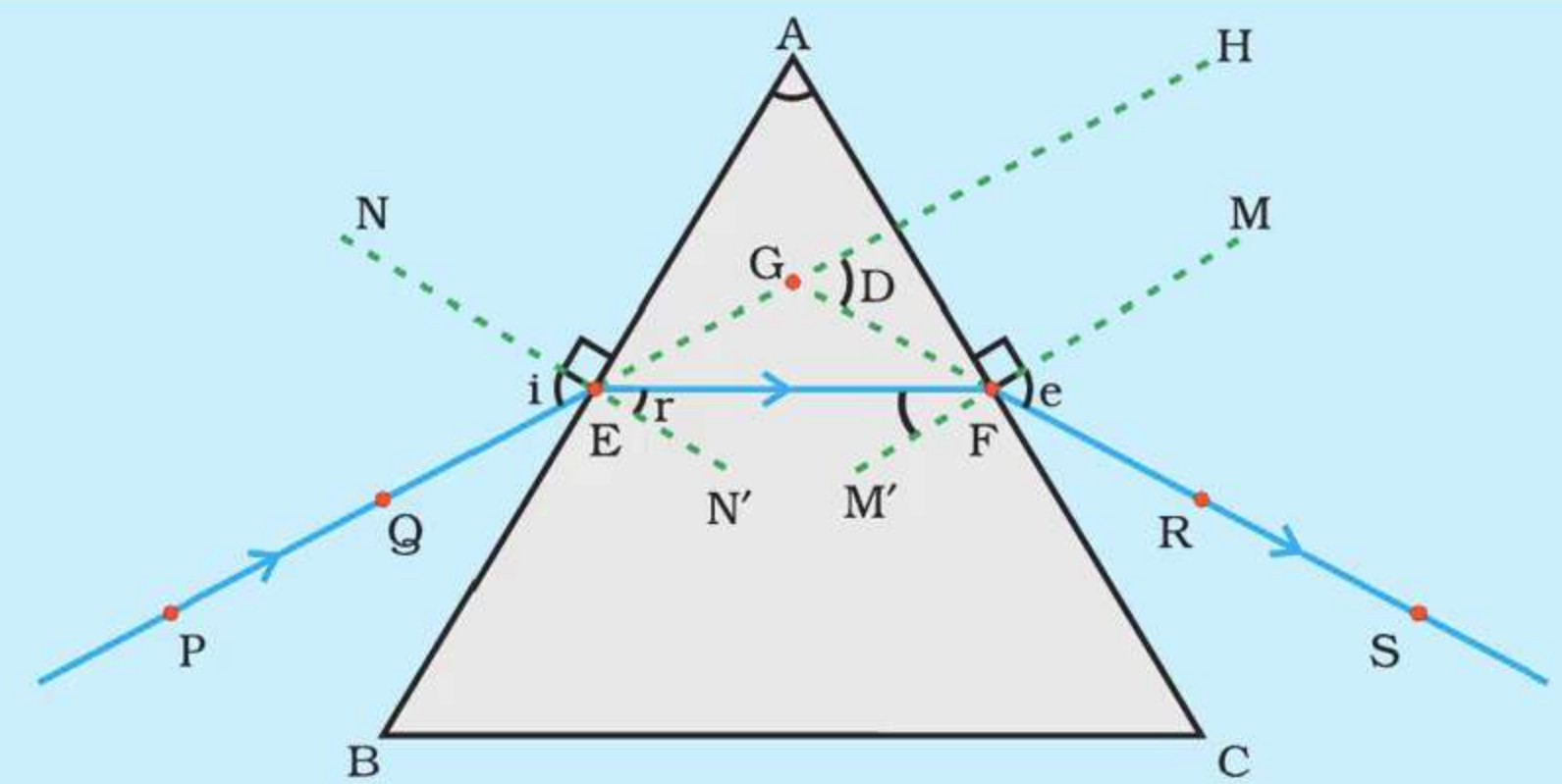


Figure 11.4 Refraction of light through a triangular glass prism

Refraction of Light Through a Prism

- Angle of Deviation ($\angle D$) :

The peculiar shape of the prism makes the emergent ray bend at an angle to the direction of the incident ray. This angle between the incident ray and emergent ray is called the angle of deviation.



PE – Incident ray

EF – Refracted ray

FS – Emergent ray

$\angle A$ – Angle of the prism

$\angle i$ – Angle of incidence

$\angle r$ – Angle of refraction

$\angle e$ – Angle of emergence

$\angle D$ – Angle of deviation

Figure 11.4 Refraction of light through a triangular glass prism

Dispersion of White Light by a Glass Prism

- Dispersion of light :

The splitting of light into its constituent colours is called dispersion.

- Formation of Spectrum :

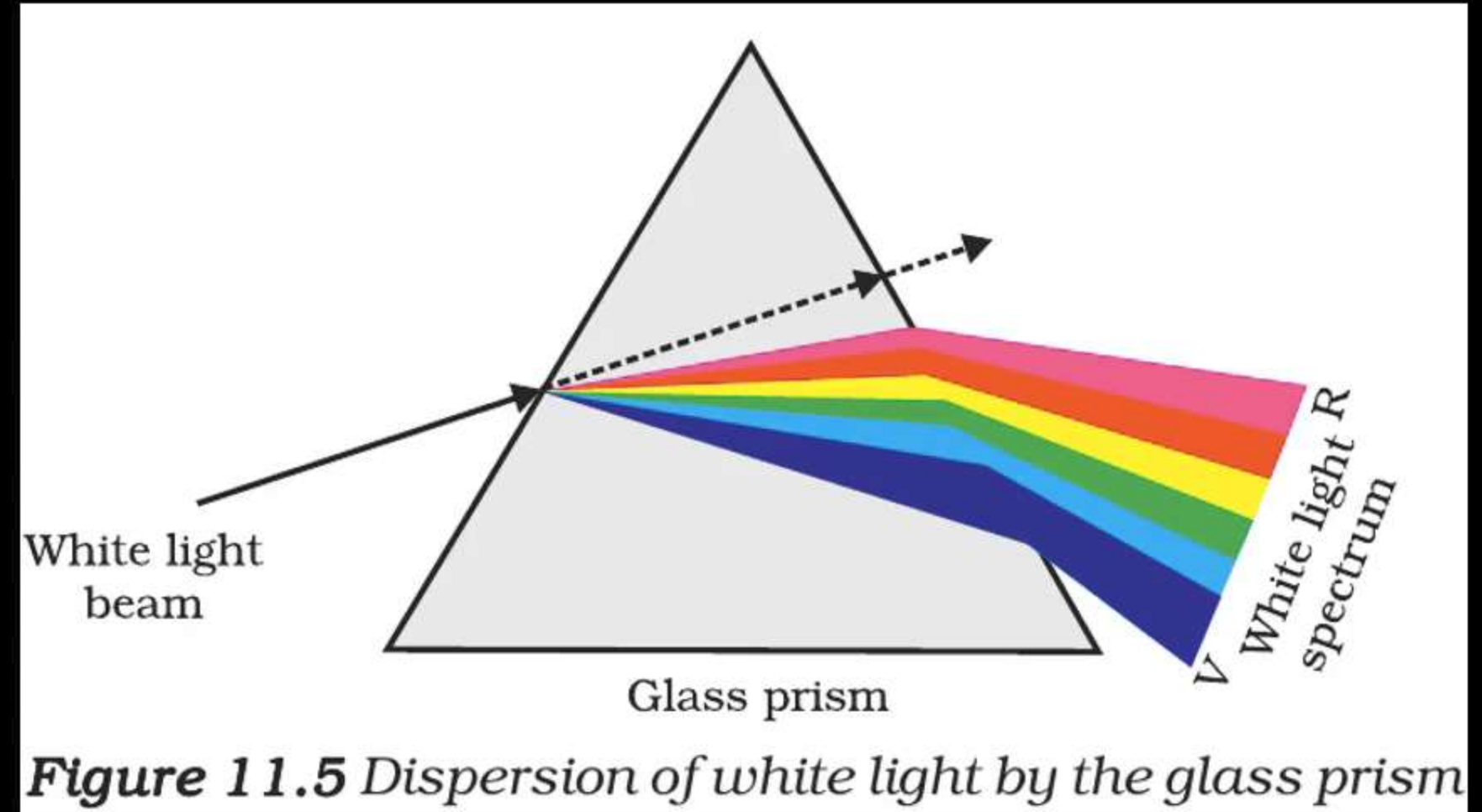
When a beam of white light is incident on a prism, it gets **dispersed into its constituent colours (VIBGYOR)**

- Reason :

Different colours of light bend through different angles as they pass through a prism. The **violet colour bends the most** and **red colour bends the least**.

- Spectrum :

The **band of colours** obtained on the screen is called the **spectrum** of light.



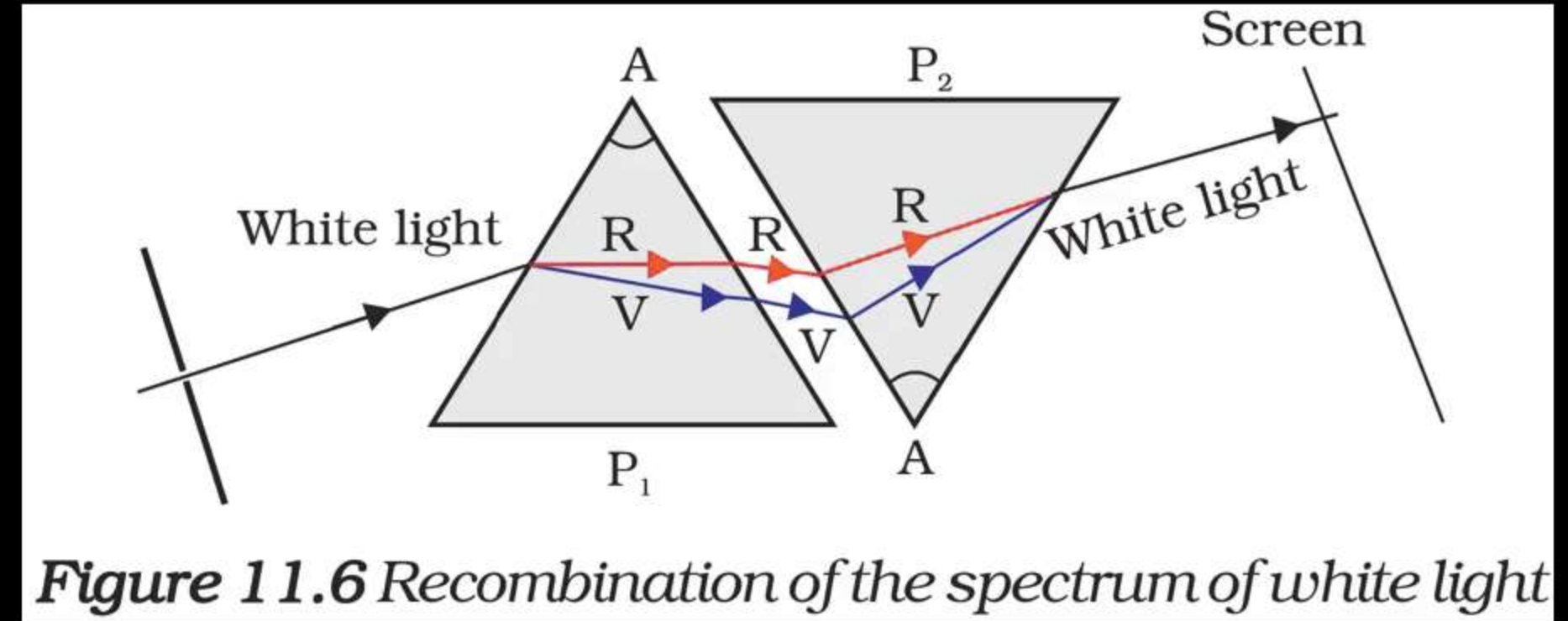
Recombination of Colours

- Recombination of Colours :

Isaac Newton used a glass prism to obtain the spectrum of sunlight. He then placed a second identical prism in an inverted position with respect to the first prism.

- Formation of White Light :

All the colours of the spectrum from the first prism when passes through the second prism. He found a beam of white light emerging from the other side of the second prism.



Rainbow

- Rainbow :

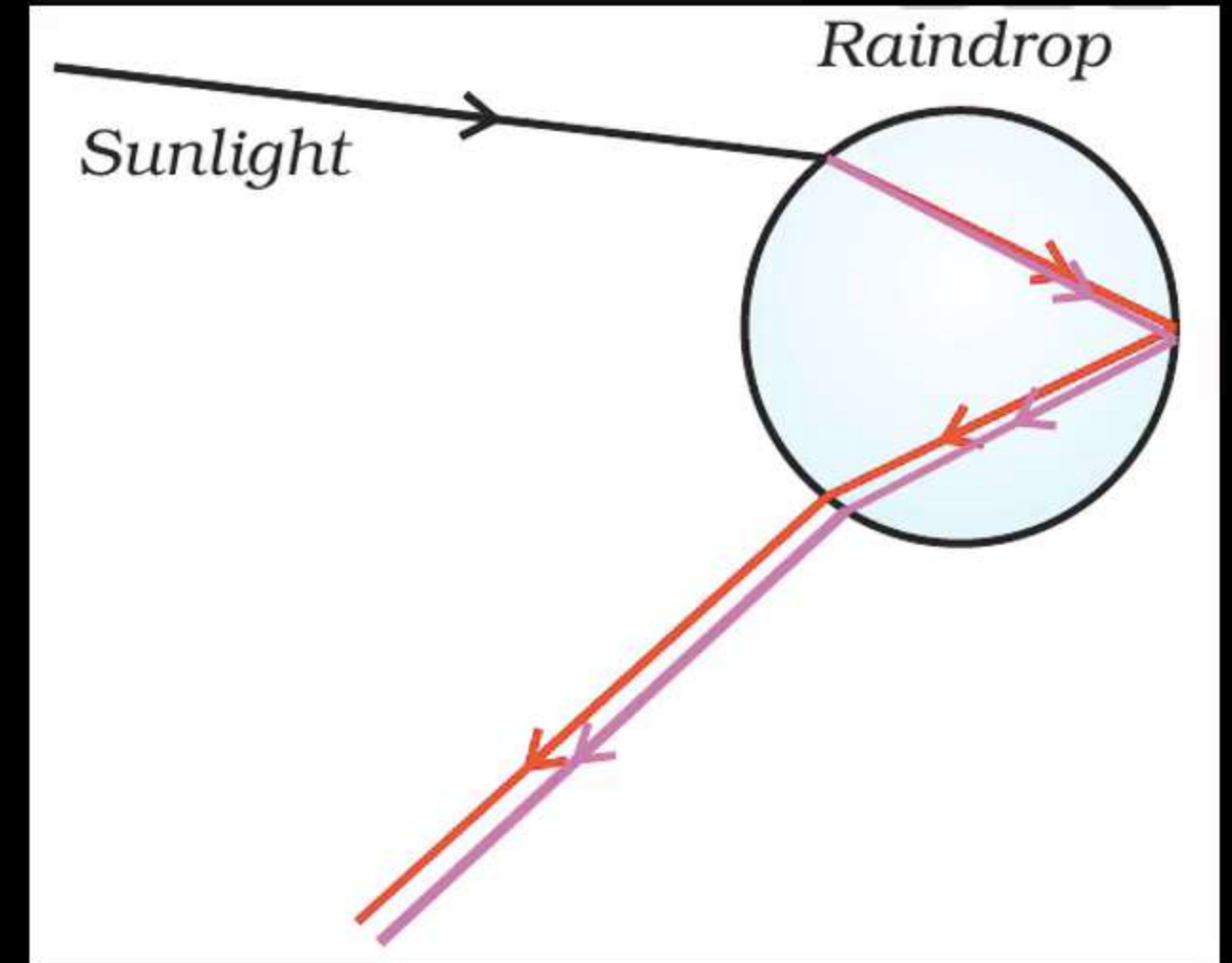
A rainbow is a **natural spectrum** appearing in the sky **after a rain shower**. A rainbow is always **formed** in a direction **opposite** to that of the **Sun**.

- Rainbow Formation :

The **water droplets** act like **small prisms**. They **refract** and **disperse** the incident sunlight, **then reflect it internally**, and **finally refract it again** when it comes out of the raindrop. Due to the **dispersion of light** and **internal reflection**, different colours reach the observer's eye.

- Other places where we can see rainbow :

You can also see a rainbow near a **waterfall** or through a **water fountain**, with the Sun behind you.



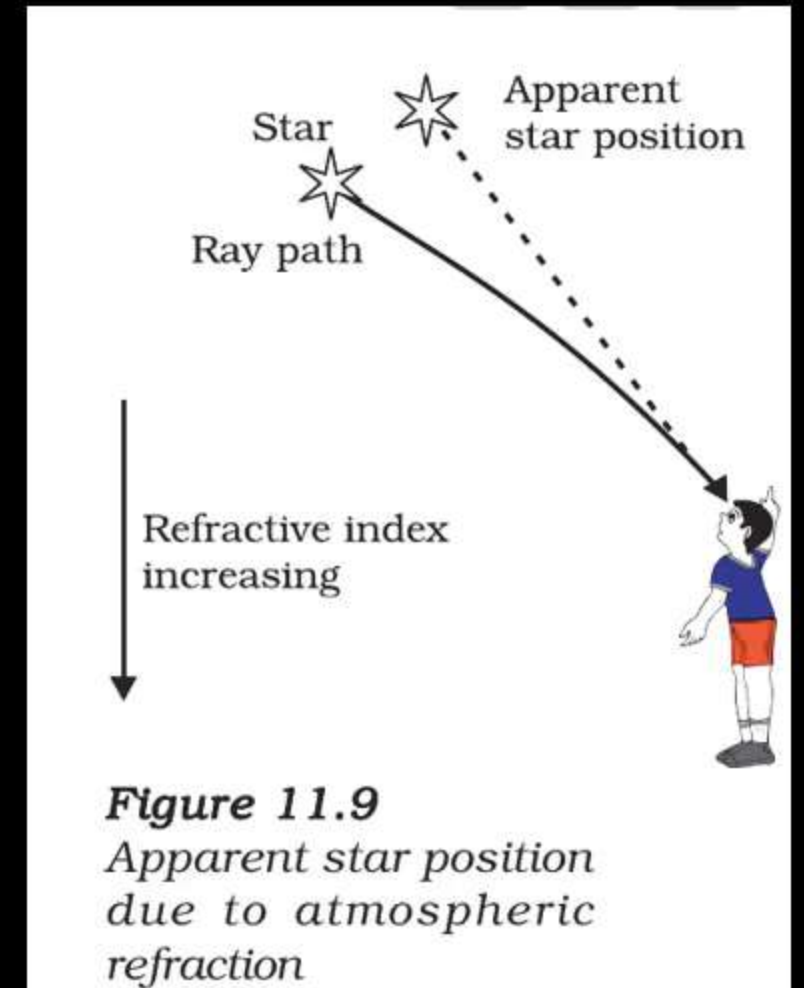
Atmospheric Refraction

- You might have observed the apparent **random wavering or flickering of objects** seen through a stream of hot air rising above a **fire**.
- The **hotter air is lighter (less dense) than the cooler air** above it, and has a **refractive index slightly less than that of the cooler air**.
- Since the **physical conditions** of the refracting medium (air) are **not stationary**, the **apparent position** of the object, as seen through the hot air, **fluctuates**.
- This wavering is thus an effect of **atmospheric refraction** (refraction of light by the earth's atmosphere) on a small scale in our local environment.



Twinkling of Stars

- The **twinkling of stars** is due to the **atmospheric refraction** of light.
- Since the **atmosphere bends starlight towards the normal**, the **star appears slightly higher (above) than its actual position** when viewed near the horizon.
- Since the **stars are very distant**, they approximate **point-sized sources of light**.
- Since the **physical conditions of the earth's atmosphere are not stationary**. The **refractive index of air is not constant** and changes with the change in density.
- This causes the **apparent position of the star to change slightly and rapidly** and the **amount of starlight entering the eye flickers**- the star sometimes appears **brighter**, and at some other time, **fainter**, which is the twinkling effect.



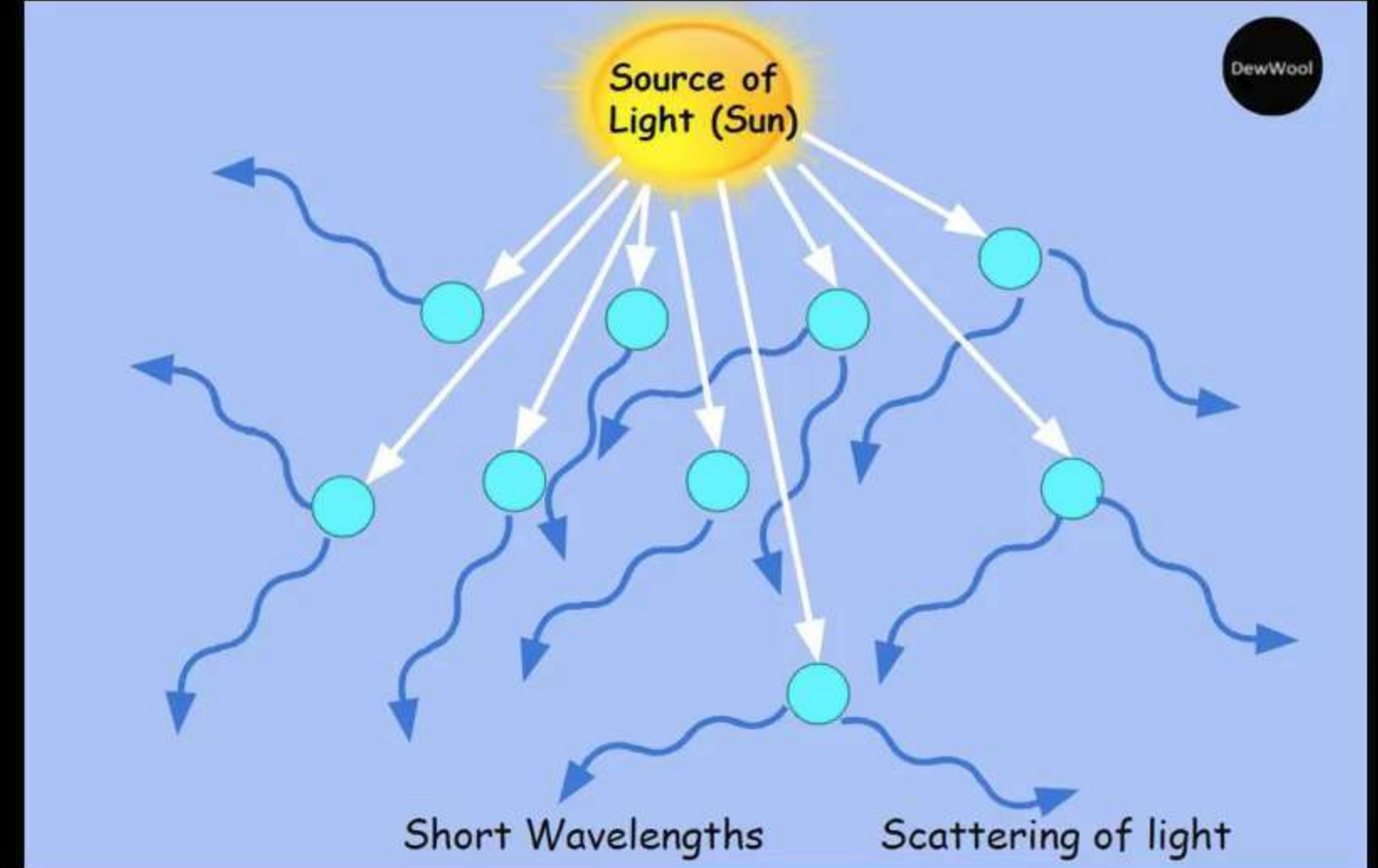
Twinkling of Stars

- **Planets are closer to the Earth** than stars, and hence they appear as **extended sources of light**.
- If we consider a **planet as a collection of a large number of point-sized sources of light**, the **total variation in the amount of light entering our eye** from all the individual point-sized sources will **average out to zero**, thereby **nullifying the twinkling effect**.

Scattering of Light and Tyndall Effect

What is Scattering of Light?

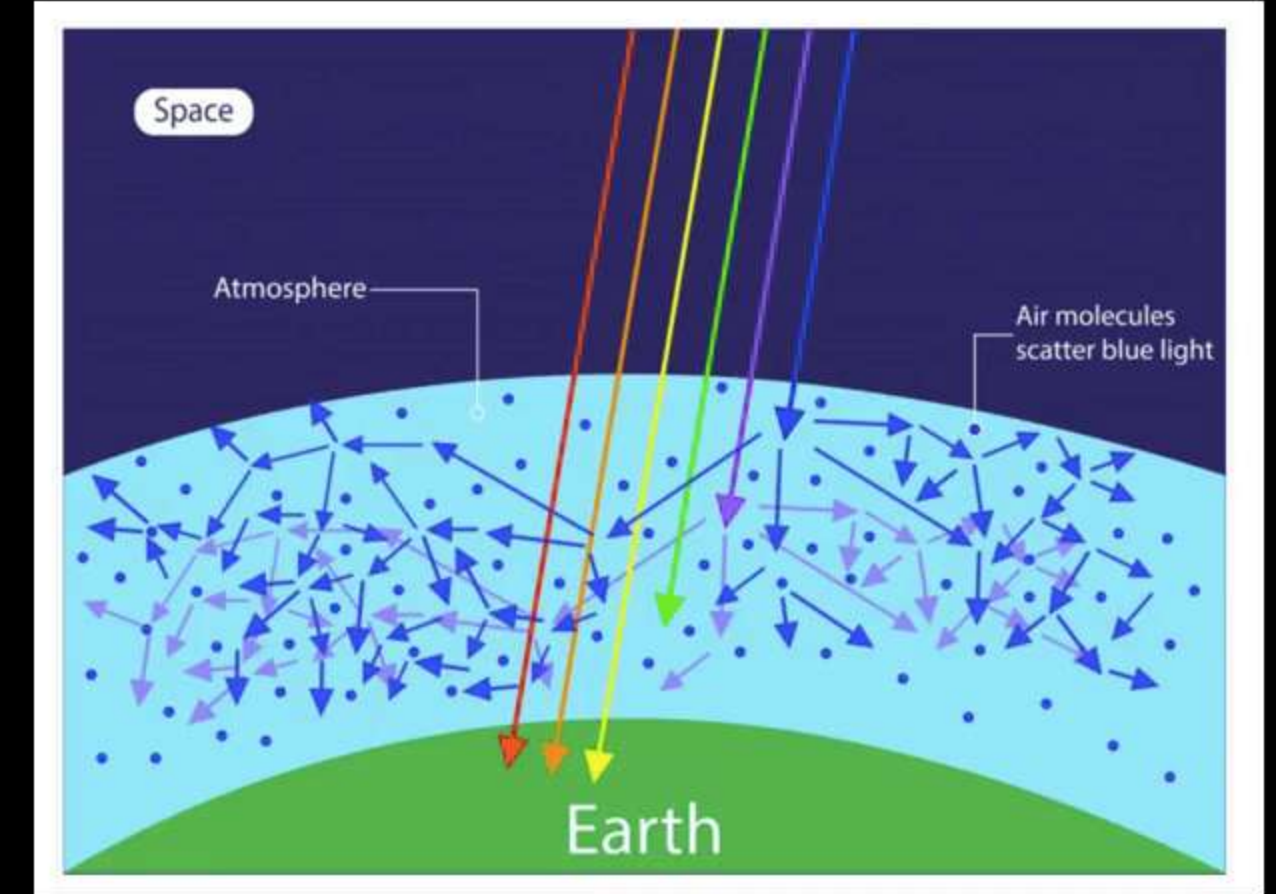
- Light **reflected diffusely** (reflect in all directions) when it strikes very small particles like air molecules or dust.
- This phenomenon is called **scattering**.
- **Shorter wavelengths** (e.g., blue) scatter more than longer ones (e.g., red).
- Explains natural effects like **blue sky** and **reddish sunset**.



Scattering of Light and Tyndall Effect

Why is the Sky Blue?

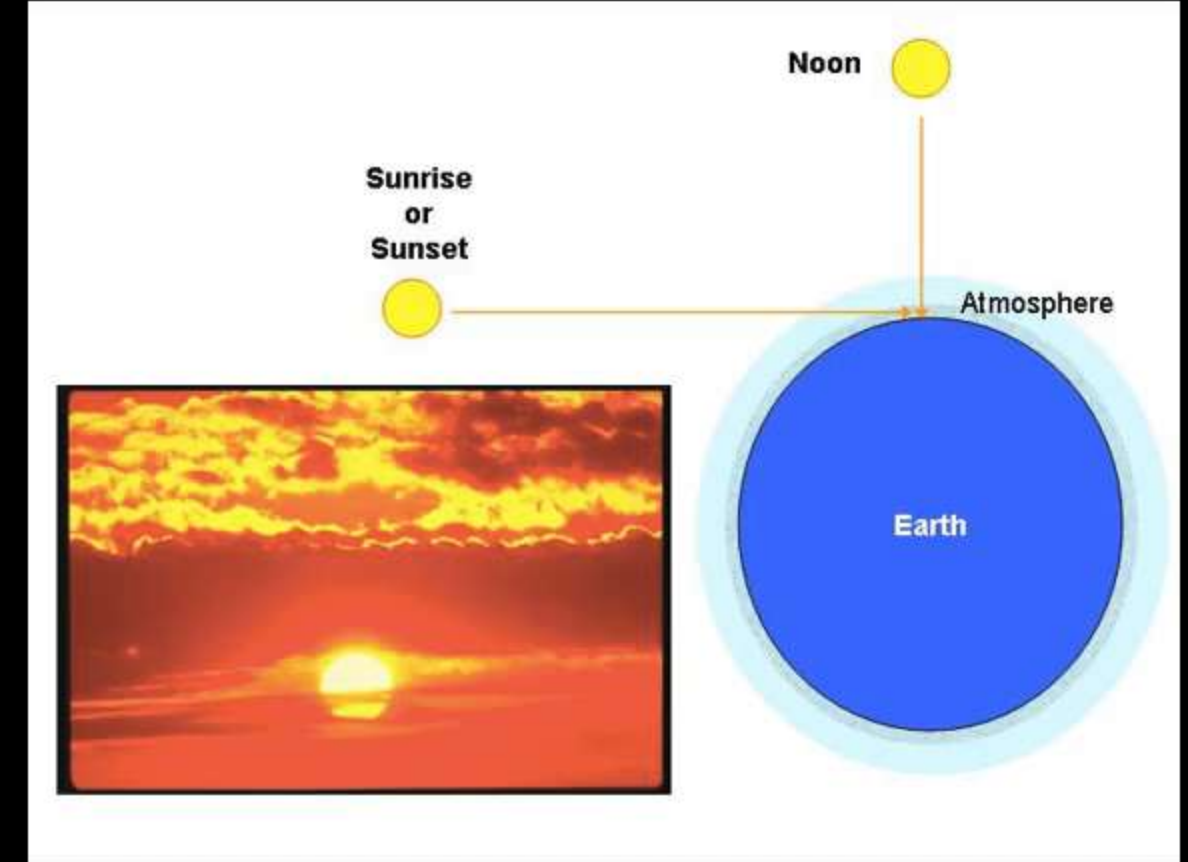
- Sunlight contains all visible wavelengths.
- **Blue light** is scattered more due to its shorter wavelength.
- Our eyes are more sensitive to blue than violet.
- So, the sky appears **blue** to us.



Scattering of Light and Tyndall Effect

Why is the Sun Red at Sunrise and Sunset?

- At sunrise/sunset, sunlight travels a **longer distance** in the atmosphere.
- **Shorter wavelengths (blue)** get scattered out.
- **Red/orange light** reaches our eyes.
- Hence, the Sun appears **reddish**.



Scattering of Light and Tyndall Effect

Tyndall Effect

- Tyndall effect is the scattering of light by colloidal particles.
- It makes the **path of the beam visible**.
- **Observed in:** fog, dusty rooms, forest mist.
- **Not seen** in pure air or true solutions.

